Classification and Statistical Evaluation with ReliableML: Metrics, Tests, and an Empirical Brier Decomposition

Jabulani Monchwe
Applied Business Analytics and Decision Making
North-West University
IDEA-Research Club Chairperson
Johannesburg, South Africa
monchwej.25@gmail.com

Martin Chanza
Applied Business Analytics and Decision Making
North-West University
Mafikeng, South Africa
martin.chanza@nwu.ac.za

Olorato Motitswane
Eduvos Midrand Campus – IT Faculty Data Science, South Africa
IDEA-Research Club
oloratomotitswane@gmail.com

Mfundo Monchwe
Eduvos Midrand Campus – IT Faculty Data Science, South Africa
IDEA-Research Club
diditmfundo@gmail.com

***Abstract*—Reliable classification evaluation requires more than a single accuracy statistic. This paper documents the labeled evaluation methods implemented in ReliableML: confusion-matrix metrics including multiclass Matthews correlation coefficient (MCC), feature–target and train–test ANOVA/MANOVA/chi-square comparisons, and an empirical repeated-stratified-cross-validation Brier decomposition. We give estimands, equations, interpretation, and assumptions, including the conditions under which MANOVA is unavailable. These outputs describe finite samples and cannot replace an independent evaluation design, calibration protocol, or correction for multiplicity. The Brier decomposition is explicitly an empirical one-hot prediction-error decomposition, not a population estimate of irreducible noise.**

***Index Terms*—classification, model evaluation, MCC, ANOVA, MANOVA, Brier score, cross-validation**

I. Introduction

Metric selection affects how classifier quality is communicated. Accuracy can obscure poor minority-class performance, while tests of feature distributions do not establish predictive utility. ReliableML keeps classification outcomes, distribution comparisons, statistical association tests, and cross-validated probability error conceptually separate. This manuscript describes the implemented computations and does not claim that their inclusion improves model performance.

II. Confusion-matrix estimands

For C classes, let M_{ij} count observations with true class i and predicted class j, $t_i = \sum_j M_{ij}$, $p_i = \sum_j M_{ji}$, $c = \sum_i M_{ii}$, and $s = \sum_{ij} M_{ij}$. Accuracy is c/s. For each class i, one-vs-rest precision, recall, specificity, and F1 are computed from $TP_i = M_{ii}$, $FN_i = t_i - TP_i$, $FP_i = p_i - TP_i$, and $TN_i = s - TP_i - FN_i - FP_i$:

$$\mathrm{Prec}_i = \frac{TP_i}{TP_i + FP_i},$$
$$\mathrm{Rec}_i = \frac{TP_i}{TP_i + FN_i},$$
$$\mathrm{Spec}_i = \frac{TN_i}{TN_i + FP_i}.$$
$$F1_i = \frac{2\,\mathrm{Prec}_i\,\mathrm{Rec}_i}{\mathrm{Prec}_i + \mathrm{Rec}_i}.$$

Zero denominators are handled by the implementation's ratio convention. Macro averages weight each observed class equally; support-weighted averages weight by true class count. Balanced accuracy is the macro-average recall.

Define $A = s^2 - \sum_i p_i^2$ and $B = s^2 - \sum_i t_i^2$. The multiclass MCC implemented by ReliableML [4] is

$$\mathrm{MCC} = \frac{cs - \sum_i t_i p_i}{\sqrt{AB}}.$$

It is a correlation-like summary that incorporates all cells and is useful when class supports differ [1], [2]. When the denominator is zero, the implementation returns zero. MCC remains only one summary; the confusion matrix, classwise rates, and supports should be reported alongside it.

III. Statistical comparisons

A. Numeric feature–target and cohort tests

For numeric feature values x_{ig} in G groups, one-way ANOVA tests equality of group means [3]. Its statistic is

$F = MS_B/MS_W$, where $SS_B = \sum_g n_g(\bar{x}_g - \bar{x})^2$, $SS_W = \sum_g \sum_i (x_{ig} - \bar{x}_g)^2$, and mean squares divide by $G - 1$ and $N - G$, respectively. ReliableML reports the F statistic, p-value, degrees of freedom, group sizes, and exclusions. Independence, approximate within-group normality, and equal variances are not automatically verified.

For multiple numeric outcomes, MANOVA tests whether group mean vectors differ jointly. The implementation uses a statsmodels multivariate linear model and returns an unavailable status if the sample is insufficient or residual covariance is rank-deficient/singular. Complete rows are required for the joint outcome vector. A singular result is not silently converted into evidence of no difference.

B. Categorical association

For a contingency table with observed counts O_{ab}, expected counts E_{ab}, the Pearson statistic is

$$\chi^2 = \sum_{a,b} \frac{(O_{ab} - E_{ab})^2}{E_{ab}}, \qquad \mathrm{df} = (A-1)(B-1).$$

It is used for categorical feature–target association and train–test cohort comparison. The report includes expected counts and warns when cells have expected count below five. Row independence and adequate expected frequencies are required for the usual asymptotic reference distribution.

These tests address whether observed means, multivariate means, or category frequencies differ; they do not quantify predictive importance or causal effects. Many features imply multiple tests, so false-discovery control and prespecified hypotheses should be considered by downstream researchers.

IV. Empirical repeated-CV Brier decomposition

For row i, let e_{y_i} be its one-hot target vector, and let $p_i^{(r)}$ be its out-of-fold probability vector on repeat r. The repeated-CV mean prediction $\bar{p}_i = R^{-1} \sum_r p_i^{(r)}$ gives empirical components

$$\widehat{B} = \frac{1}{n} \sum_i \|\bar{p}_i - e_{y_i}\|_2^2, \quad \widehat{V} = \frac{1}{n} \sum_i \frac{1}{R} \sum_r \|p_i^{(r)} - \bar{p}_i\|_2^2,$$

$$\widehat{E} = \frac{1}{n} \sum_i \frac{1}{R} \sum_r \|p_i^{(r)} - e_{y_i}\|_2^2.$$

The implementation reports these empirical quantities from repeated stratified cross-validation. Probability outputs are used when available; otherwise hard predictions are one-hot encoded. The fold count is bounded by the smallest observed class size. This finite-sample decomposition depends on the selected folds, repeats, estimator, and labels; it is not an estimate of population irreducible noise and does not remove selection bias if model choices were made using the same data.

V. Evaluation design and interpretation

The test data must be separated from all fitting, preprocessing, feature selection, resampling, threshold setting, and hyperparameter selection. Choose a split design aligned with deployment (random, grouped, spatial, temporal, or external-site). For selected-model claims, use nested or untouched outer validation. Report label prevalence, per-class support, confusion counts, calibration, uncertainty, and the chosen primary metric. Statistical p-values are not effect sizes and should be interpreted with sample size and multiplicity in view.

VI. Conclusion

ReliableML combines classwise metrics, MCC, statistical cohort tests, and repeated-CV Brier summaries. Each quantity answers a different question and inherits its own sampling assumptions. The package provides diagnostics; valid generalization claims still depend on a prospectively chosen evaluation design and independent data.

References

[1] J. Gorodkin, "Comparing two K-category assignments by a K-category correlation coefficient," *Computational Biology and Chemistry*, vol. 28, no. 5–6, pp. 367–374, 2004.

[2] B. W. Matthews, "Comparison of the predicted and observed secondary structure of T4 phage lysozyme," *Biochimica et Biophysica Acta*, vol. 405, no. 2, pp. 442–451, 1975.

[3] R. A. Fisher, *Statistical Methods for Research Workers*. Edinburgh: Oliver and Boyd, 1925.

[4] J. Monchwe, O. Motitswane, and M. Monchwe, *ReliableML*, software, version 0.1.1, 2026. https://github.com/Jabualani/ML-Reliabilty.